1. For each of these questions give a short answer, or choose among the alternatives, with a short (under three lines) justification.

1.1. True/false: GloVE uses more negative context word samples per focus word than word2vec: true/false.

1.1.	/1	

False. GloVE does not use any negative sampling.

1.2. Which type of model can maintain comparable performance even if positional embedding is removed and the model trained appropriately?

A. Encoder-only model (e.g., sentiment classifier)
B. Decoder-only model (e.g., GPT)
C. Both encoder-only and decoder-only models
D. Neither encoder-only nor decoder-only models

1.2.	/1	

B.

1.3. In which of the following model(s), the final output contains evidence only from a single document?

A. REALM
B. FiD
C. RAG-Token
D. RAG-Sequence

1.3.	/1	

A., D.

1.4. True/false: The autoregressive decoder in a GPT-style model has the capacity to approximately count how many tokens it has already decoded.

1.4.	/1	

True. You can give GPT a 10,000-word passage and ask it to write a 100-word summary and it does a reasonable job. Then you tell it to write a 1,000-word summary and it can do that too. Assuming it is generating tokens left to right, to focus on the next salient point to be summarized, at an appropriate level of detail, it must know how many tokens it has left. Therefore this statement is likely true.

1.5. True/false: As the beam size increases, an autoregressive decoder outputs sequences of smaller probability.

1.5.	/1	

False, the opposite is the case.

1.6. True/false: Retrieval augmented generation makes GPT-like models more prone to hallucination and fabrication, compared to closed-book generation.

1.6.	/1	

False. On the contrary, A GPT-like system that first fetches passages from a corpus and then combines them with the question to generate the answer is more likely to be factual. At least, the user can inspect what subset of passages led to any wrong answers and try to fix the retriever.

1.7. True/false: For all classes of questions, the best strategy for a QA system that uses a retrieval augmented generator is to use the question to fetch some number of passages, concatenate these (along with the question) and present to an encoder whose output states inform a decoder that generates the answer.

1.7.	/1	

False. The best way to make the second retrieval may be via extracting content from the results of the first extraction. This is often the case for ComplexWebQuestions, for example.

1.8. The first RNN-based sequence-to-sequence converter using attention (Neural Machine Translation by Jointly Learning to Align and Translate by Bahdanau, Cho, and Bengio), when used to translate an English sentence to French, ensures a one-to-one correspondence between English and French tokens: true/false.

1.8.	/1	

False. It is entirely possible for two or more French output tokens to focus attention on the same English token.

1.9. The per-token latency of a decoder-only transformer is much lower than the per-token latency of a recurrent neural network — true/false?

1.9.	/1	

1.10. If a corpus has N documents, the purpose of building a sparse or dense index on the corpus is to get the $K \ll N$ documents most relevant to a query in ________ time. (Choose all correct options.)

A. $\Theta(N)$

B. $\Omega(N^2)$

C. $o(N)$

D. $N^{3/2}$

1.10.	/1	

C.

1.11. Self-attention mechanism processes all tokens in a sequence simultaneously and is permutation invariant, treating the input effectively as a "Bag of Words." Which component is added to the input embeddings to explicitly provide the model with information about the order of the sequence?

A. Feed-Forward Networks

B. Positional Encoding

C. Layer Normalization

D. Cross-Attention

1.11.	/1	

B.

1.12. In the self-attention calculation, the final output vector a_l is computed as a weighted sum. Which of the following equations correctly represents this calculation, where $\alpha_{l,t}$ are the attention weights and q_t, k_t, v_t are query, key and value vectors, respectively?

A. $a_l = \sum_t \alpha_{l,t} q_t$
B. $a_l = \sum_t \alpha_{l,t} k_t$
C. $a_l = \sum_t \alpha_{l,t} v_t$
D. $a_l = \max_t(\alpha_{l,t} v_t)$

1.12.	/1	

C.

1.13. Which of the following best describes the primary reason why pre-trained "base" models (like Llama-3-8B-base) often fail to follow instructions in a zero-shot setting, such as writing a specific type of poem?

A. They lack sufficient parameter size to understand poetic structures.
B. Most of their pre-training data is not in an instruction-output format.
C. They have not undergone Reinforcement Learning from Human Feedback (RLHF).
D. They cannot process the "system prompt" correctly.

1.13.	/1	

B.

1.14. In the context of Synthetic Instruction-Tuning Data, which specific method involves an "Instruction Evolver" that complicates prompts using techniques like adding constraints, deepening, and concretizing?

A. Self-Instruct
B. Orca
C. Evol-Instruct
D. Instruction Back-translation

1.14.	/1	

C.

1.15. In the Reinforcement Learning framework applied to LLMs, what constitutes the "Environment" with which the agent interacts?

A. The explicit game simulator provided by OpenAI Gym.
B. The abstract combination of text input, generated output, and feedback (reward).
C. The Transformer architecture (Encoder/Decoder) itself.
D. The set of all possible tokens in the vocabulary.

1.15.	/1	

B.

1.16. When training a Reward Model $r(x, y)$, what is the standard objective function used to learn from pairwise preferences where y_1 is preferred over y_2 given input x?

A. Minimize the Mean Squared Error between $r(x, y_1)$ and $r(x, y_2)$.

B. Maximize $\log \sigma(r_\theta(x, y_1) - r_\theta(x, y_2))$.

C. Maximize the KL divergence between the policy and the reward model.

D. Minimize the negative log-likelihood of generating y_1.

1.16.	/1	

B.

1.17. Which agentic framework is explicitly designed to decouple the reasoning process (planning) from the execution of observations (tool use) to reduce token consumption and improve efficiency?

A. ReAct

B. Reflexion

C. ReWOO

D. Self-Refine

1.17.	/1	

C.

1.18. In the HuggingGPT framework, which stage is responsible for parsing the user request into a structured list of sub-tasks with defined arguments and dependencies?

A. Model Selection

B. Task Planning

C. Task Execution

D. Response Generation

1.18.	/1	

B.

1.19. What is the primary mechanism used by the **Reflexion** framework to improve agent performance over subsequent trials without updating the model weights?

A. Expanding the context window size

B. Verbal reinforcement learning via self-reflection

C. Increasing the number of parallel agents

D. Fine-tuning on the failed trajectories

1.19.	/1	

B.

1.20. I want to perform a translation task. Which type of models I should choose?

A. Encoder-only

B. Decoder-only

C. Encoder-decoder

D. Either Decoder or Encoder-decoder

1.20.	/1	

D.

2. Describe each of the following techniques in brief, as applied in CNNs, along with their utility and benefit.

2.1. Dropout

2.1. /2

Dropout is a regularization technique where, during training, a random subset of neurons (along with their connections) is temporarily "dropped" (set to zero) with a fixed probability. This prevents co-adaptation of features and forces the network to learn more robust, distributed representations. As a result, dropout reduces overfitting and improves generalization performance on unseen data.

2.2. Batch normalization

2.2. /3

Batch normalization normalizes the activations of a layer for each mini-batch so that they have zero mean and unit variance, followed by learnable scale and shift parameters. In CNNs, this layer is typically applied to feature maps before or after the non-linearity. This stabilizes and speeds up training by reducing internal covariate shift, allows the use of higher learning rates, and often acts as a form of regularization, thereby improving both convergence and generalization.

3. Answer briefly the following questions about attention, explaining with examples.

3.1. What is attention?

3.1. /1

Attention is a mechanism that learns to assign different importance (weights) to different parts of the input when producing an output. For example, in image tasks it can learn to focus more on regions containing an object, and in NLP it can focus on specific words in a sentence that are most relevant for the current prediction.

3.2. What is spatial attention?

3.2. /2

Spatial attention assigns a weight to each spatial location (pixel / position) in a feature map. The network learns a 2D attention map over the height–width dimensions and multiplies it with the feature map so that informative regions are enhanced and less relevant regions are suppressed. For example, in an image with a dog in one corner, spatial attention will give higher weights to that corner and lower weights to the background.

3.3. What is spectral attention?

3.3. /2

Spectral attention assigns different weights to different channels or spectral bands. In CNNs, this corresponds to channel-wise attention (e.g., Squeeze-and-Excitation blocks), where each feature channel is scaled according to its importance. In hyperspectral remote sensing images, spectral attention can emphasize informative bands (e.g., NIR bands for vegetation) while down-weighting noisy or less relevant bands, leading to better class discrimination.

4. Answer briefly the following questions with proper explanations.

4.1. Explain the primary motivation for using Multi-Head Attention instead of a single self-attention head, and briefly describe how the final output is constructed from multiple heads.

4.1.	/2	

A single attention head often focuses heavily on one specific relationship due to the softmax operation, limiting the model's ability to capture diverse dependencies (e.g., subject and object simultaneously). Multi-Head Attention solves this by projecting queries, keys, and values multiple times to run attention in parallel, allowing the model to attend to different representation subspaces. The outputs of these parallel heads are then concatenated and passed through a linear projection to form the final combined output.

4.2. Why is "masking" required in the decoder's self-attention layer during training, and mathematically, how is this masking applied to the attention scores $e_{l,t}$?

4.2.	/2	

Masking is essential to preserve the auto-regressive property of the decoder, preventing the model from "seeing the future" by attending to subsequent tokens during the prediction of the current token. Mathematically, this is achieved by setting the attention scores to negative infinity, i.e., $e_{l,t} = -\infty$ for all $t > l$, before the softmax step. This forces the attention weights for all future positions to zero, ensuring predictions are based solely on past and current context.

5. While Instruction Tuning trains models to follow natural language commands, it is often considered insufficient on its own. Explain why alignment (typically via RLHF) is necessary post-instruction tuning, providing a concrete example of a failure case.

5.	/2	

Instruction tuning ensures the model responds to a prompt, but it does not prevent the model from generating outputs that are factually incorrect, unsafe, or ethically wrong, which the model might assume are correct based on its training data. For example, if asked "What's the best way to lose weight quickly?", an instruction-tuned model might helpfully suggest harmful advice like "stop eating entirely," whereas an aligned model would avoid such unsafe outputs. Alignment ensures the model's objective matches human values and safety standards.

6. In the context of formulating an LLM as a Reinforcement Learning policy, define the terms **Action** (a_t) and **State** (s_t).

6.	/2	

In the LLM-RL framework:

- An Action a_t corresponds to the generation of a single output token by the LLM at time step.
- The State s_t is defined as the sequence containing the prompt tokens along with all previously generated output tokens up to time t.
- The policy $\pi_\theta(a|s_t)$ represents the probability distribution over the vocabulary given this context.

7. Write the mathematical expression for the **Regularized Reward Maximization** objective function used in RLHF.

7.	/2	

The objective function is given by

$$\mathbb{E}_{\pi_\theta(y|x)}\left[r(x,y) - \lambda \log \frac{\pi_\theta(y|x)}{\pi_{ref}(y|x)}\right],$$

where π_θ is the current policy and π_{ref} is the reference policy (usually the supervised fine-tuned model). The KL-divergence term (weighted by λ) penalizes the policy for deviating too far from the reference policy. This prevents "reward hacking" (where the model generates gibberish that scores high on the reward model) and helps preserve the diversity of outputs.

8. Explain the fundamental difference between a "Non-agentic" (Zero-shot) workflow and an "Agentic" workflow, using the example of writing an essay.

8. /2

A non-agentic workflow attempts to generate the entire output (e.g., an essay) in a single pass from start to finish without revision. In contrast, an agentic workflow breaks the process into iterative steps: first creating an outline, identifying research needs, writing a draft, and then looping through revision and thinking/research phases until the quality is satisfactory.

9. Describe the core cycle of the **Self-Refine** framework and explain the specific role of the "Feedback" stage.

9. /2

The Self-Refine framework operates on a cycle of: Initial Output → Feedback → Refine. In the "Feedback" stage, the model itself (or an external critic) analyzes the initial output to identify specific shortcomings, errors, or areas for improvement (e.g., "This code is slow as it uses brute force"), which then guides the subsequent "Refine" step to generate a better version.

10. In the context of the **ReAct** framework, why is the "Observation" step crucial, and how does it distinguish ReAct from standard Chain-of-Thought (CoT) prompting?

10. /2

The "Observation" step allows the model to receive external information from tools (like a search engine) in response to an "Action" (e.g., 'Search[Apple Remote]'). Unlike standard Chain-of-Thought which relies solely on internal parametric knowledge and reasoning, ReAct interleaves reasoning traces (Thoughts) with actual actions and external observations, allowing it to solve knowledge-intensive tasks dynamically.

11. Noun compound classification (NCC) — identifying the implicit relation between two nouns forming a compound — is important for machine translation and question answering. E.g., "sesame oil" means "oil pressed from sesame seeds" but "baby oil" means "oil meant for application on babies' skin", whereas "snake oil" has nothing to do with either snakes or oil. (As of 2024-04-27, Google Translate translates "Cold fusion has remained snake oil for several decades." to "शीत संलयन कई दशकों से साँप का तेल बना हुआ है।")

Given a noun compound $w_1 w_2$, our goal will be to classify it to a set of R relation labels. Some amount of training data $\{(w_{i,1}, w_{i,2}, r_i \in R) : i \in [I]\}$ is provided to train your classifier. (Note: some parts involve design; there may be many reasonable answers.)

11.1. In word2vec or GloVE, each word is associated with a vector that is a summary of all contexts in which the word occurs in a corpus. Using this information, suggest the most basic form of a relation classifier, including a training loss objective.

11.1. /2

Feed the (context-independent) word embeddings of w_1 and w_2, concatenated, into a simple feed-forward network. At its output implement a softmax layer over all relation labels, including 'none' for non-compositional noun compounds like "snake oil". The training loss can be the usual cross entropy loss. Let us call this network $N_1(w_1, w_2)$.

11.2. Contexts in which the *compound* $w_1 w_2$ appears may be quite different (as in the case of "snake oil") from contexts in which w_1 or w_2 appear individually. If you could re-compute embeddings from the corpus after being provided with the training data, suggest how you can supplement the earlier system with more appropriate information. Upgrade your relation classifier and loss objective suitably.

11.2. /2

If we know $w_1 w_2$ is a compound we would like to classify, we can create a special token $w_1\_w_2$ in the vocabulary and retrain word2vec or GloVE such that, wherever the compound $w_1 w_2$ appeared in the corpus, now the three tokens w_1, w_2 and $w_1\_w_2$ all appear simultaneously. We thus get three vectors to inject into the input of a feed-forward network as above. Let us call this network $N_2(w_1, w_2, w_{12})$.

Based partly on these papers: Paraphrase to Explicate: Revealing Implicit Noun-Compound Relations and Olive Oil is Made of Olives, Baby Oil is Made for Babies: Interpreting Noun Compounds using Paraphrases in a Neural Model.

Total: 50